

Mustafa Sultan

AI Engineer | Machine Learning & Deep Learning | LLM Applications

Cairo, Egypt | +20 1012442284 | mustafa_sultan03@outlook.com

linkedin.com/in/mustafa-sultan-34251b227 | github.com/mustafa200312

mustafa200312.github.io | scholar.google.com/citations?user=0GaeCUYAAAAJ

Professional Summary

AI Engineer and recent Artificial Intelligence graduate with hands-on experience in machine learning, deep learning, LLM applications, agentic workflows, backend APIs, NLP, and computer vision. Built practical AI systems through internships, research, and academic projects, and contributed to a peer-reviewed publication.

Technical Skills

Programming:	Python, SQL, C++, C#
AI/ML:	PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Machine Learning, Deep Learning, NLP, Computer Vision
Generative AI:	LLM Integration, LangGraph, LangChain, Agentic AI, Multi-Agent Systems, RAG, Hybrid Search, Reranking, Embeddings, Retrieval, Prompt Engineering, Tool Calling
Backend & Data:	Django, Django Ninja, FastAPI, REST APIs, PostgreSQL, SQL Databases, Pandas, NumPy, Pydantic, API Integration
Software Engineering:	Git, GitHub, OOP, Data Structures and Algorithms, Version Control
Additional:	Qiskit, Quantum Optimization, Competitive Programming

Education

B.Sc. in Artificial Intelligence <i>Nile University</i>	Oct 2022 - Jun 2026 <i>Egypt</i>
- GPA: 3.82 / 4.00; Dean's List: Fall 2024. - Completed and passed all academic requirements; official graduation documentation pending.	

Experience

AI Engineer Intern <i>Crossworkers Egypt</i>	Jul 2025 - Sep 2025 <i>Cairo, Egypt</i>
- Built an AI lead-generation pipeline that analyzed email newsletters to extract job postings and company information. - Developed a machine learning model to predict employee satisfaction from historical HR data. - Created an HR chatbot for natural-language querying over SQL databases. - Integrated LLM APIs and LangGraph to orchestrate agentic workflows across backend services and live databases. - Implemented application components using Python and C# for internal AI workflows.	
Research Intern - Machine Learning Classification <i>Nile University</i>	Jun 2024 - Sep 2024 <i>Giza, Egypt</i>
- Performed preprocessing, EDA, and feature engineering on mobile traffic and POI datasets. - Implemented and evaluated classification models for socioeconomic indicator prediction. - Contributed to a peer-reviewed publication based on the project results.	
Research Intern - Quantum Optimization <i>Nile University</i>	Jun 2024 - Sep 2024 <i>Giza, Egypt</i>
Applied QAOA to unit commitment optimization problems and explored quantum optimization concepts in a research setting.	

Selected Projects

FGaiB - Multi-Agent Personal Finance Android Application	2025 - 2026
- Developed an end-to-end Android personal finance application connected to Django Ninja REST APIs and PostgreSQL. - Built LangGraph workflows that routed requests to specialized financial agents while maintaining relevant user context. - Implemented validated financial operations, confirmation steps, conversation memory, budget-risk alerts, goal tracking, and spending analysis. - Worked across the mobile interface, backend APIs, database design, and AI orchestration to turn the idea into a working product prototype. - The project ranked first in one of eight faculty graduation project committees.	

SlideTutor - AI-Powered PowerPoint Study Workspace

- Built a full-stack educational AI application that extracts and renders PowerPoint slides for interactive study.
 - Developed a FastAPI backend and React interface for slide content, images, structured elements, and slide-aware chat.
 - Implemented LangGraph summarizer and interactive agents with conversation memory and OpenAI-compatible model support.
- github.com/mustafa200312/slidetutor

AI-Maktaba AI-Digital - Arabic Book-Constrained RAG System

- Built a citation-enforced RAG system over selected Arabic classical books and texts.
- Implemented Arabic document processing, chunking, embeddings, hybrid search, reranking, and grounded answer generation.
- Combined semantic and keyword-based retrieval, then reranked retrieved passages to improve source relevance.

Multi-View Ultrasound Imaging with a Vision Foundation Model

- Developed a two-view PyTorch model using a frozen pretrained DINOv3 backbone for longitudinal and transverse ultrasound scans.
- Built three-class segmentation for background, vessel, and plaque regions, plus binary plaque classification using both views.
- Converted grayscale ultrasound images into a format compatible with DINOv3, trained with cross-entropy and Dice-based losses, handled class imbalance, and evaluated per-class Dice and F1.

MAAP - Multi-Agent System for Automatic Code Parallelization

- Built a multi-agent LLM system that transforms sequential Python and C code into parallel implementations using agents for dependency detection, task decomposition, and execution planning.
- github.com/mustafa200312/MAAP-Multi-Agentic-for-Auto-Parallelization

Publication & Achievements

- *A Multi-label Classification Framework for Socioeconomic Insights Using Mobile Traffic and POI Data.* doi.org/10.1007/978-3-031-87719-3_16
- Qualified for ECPC 2023 and 2024.
- Won 2nd Place in the Linear Algebra Course category at the 17th Undergraduate Research Forum, Nile University.

Leadership, Volunteer Work & Community Engagement

Junior Teaching Assistant <i>Nile University</i>	Oct 2023 - Jan 2024
• Supported students in programming, data structures, algorithms, data analysis, preprocessing, and feature engineering. • Explained technical concepts, helped debug code, and guided students through practical exercises.	
Junior Mentor / ICPC Community Mentor <i>Nile University ICPC Community</i>	Oct 2023 - Jun 2024
• Mentored students in competitive programming, algorithms, and problem solving. • Helped prepare weekly Codeforces problem sheets, delivered a technical session, and assisted with ECPC registration documents.	
Vice President <i>University Gaming Club</i>	Sep 2024 - May 2025
• Helped manage the campus gaming community, co-organized casual gaming sessions and small tournaments, and supported event planning and logistics.	